

Karnataka's Mathematics Performance | Policy Note

From: Team datathon_csf | Datathon 2026 | 1 August 2026

What the data shows

We looked at 13,79,087 answer sheets from the Akshara maths contest: classes 4, 5 and 6, three years, rural government schools.

Class 6 got worse at two things. Multiplication fell from 51% correct to 35%. The performance on division fell from 55% to 38%. Class 4 held steady. So the older children now score lower than the younger ones, which is backwards.

ASER 2024 and PARAKH 2024 rank Karnataka's districts the same way this assessment does, so this is not one odd test.

Three things worth knowing

1. **The problem is stepped progression in maths, not competencies standalone.** Children who can multiply get division right 45 points more often. Children who can add get subtraction right 48 points more often. Skills are complimentary, so you cannot fix division by teaching more division.
2. **Kalyana Karnataka is short of teachers and falling further behind.** Its rural government schools have 33.6 children per teacher against 20.4 elsewhere, and 139 children per school against 73. Its score gap widened from 10 to 16 points in three years. That is a resourcing problem, and resourcing can be changed.

What we recommend

1. Run a class 5 and 6 catch-up programme on multiplication and division. Start in the weakest Clusters, not across the whole state.

Owner: DSERT, delivered through Block Education Officers and the CRP cadre.

By: materials ready for Term 2, 2026-27.

Cost: low. Kalika Balavardhane already exists. This is reprinting plus one round of training.

Success: class 6 multiplication back above 45% in the next round, measured in the same villages.

2. Fix the testing gaps before trusting more year-on-year comparisons. Only 50% of eligible children were tested in the latest round, even considering rural schools within state government management only. Two districts tested nobody. Three more tested under 6%. And the three highest-scoring districts are the three that tested fewest children, making the rankings unreliable for these.

Owner: District Education Officers with District Assessment Coordinators.

By: within two assessment rounds.

Cost: low. This is logistics, not a new test.

Success: state participation well above 50%, and no district below half the state average.

3. Provision better in Kalyana Karnataka's 7 Districts: Ensure the supplies and teacher availability in these backward districts is corrected on priority.

Owner: Samagra Shiksha MIS cell, with Cluster Resource Persons in the field.

By: next planning and budget round.

Cost: Significant. We have to make PTR as per the RTE Act, so the cost is legislatively backed.

Success: a School-level dashboard in real use for one full year showing provisions made vs test scores improved.

What we are not saying

Byndoor block in Udupi scores 85% where its conditions predict 54%. This data shows things moving together, not causes. And the questions change every year, so we compare skills, never single questions, and check against two outside surveys.

Sources. Three years of the Akshara GP Maths Contest joined to UDISE+ 2022-25, Census 2011, NFHS-5 and 6, ASER 2024, PGI-D 2025-26 and PARAKH 2024. Rebuild offline with python `src/run_all.py`. Every number is listed in `claims.json`. Repository: github.com/common-del/datathon_csf